Zabed Ibne Emdad

Software Engineer | MSc Artificial Intelligence, University of Essex

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PROFESSIONAL PROFILE

Software engineer with industry experience, now completing an MSc in Artificial Intelligence focused on large language models for emergency healthcare. Builds and evaluates LLM systems end to end: retrieval-augmented generation, parameter-efficient fine-tuning (LoRA, QLoRA) with PyTorch and Hugging Face PEFT, agentic tool-use workflows, and calibration-focused evaluation (ECE, Brier score) in high-risk decision settings.

Backs this with production engineering discipline: enterprise software delivery, CI/CD automation and cloud deployment. Works comfortably across distributed, cross-functional teams, translates technical detail for non-specialist stakeholders, and mentors junior engineers as a matter of routine.

PROFESSIONAL EXPERIENCE

Software Engineer — Freelance, Kuala Lumpur February 2024 – March 2025

Independent consultant delivering client-facing products end to end.

- Built full-lifecycle mobile and web applications for clients, integrating proprietary backends and third-party services, and owned delivery from requirements gathering to release.
- Shipped LLM-powered features — content generation and structured outputs via generative AI APIs — built with LangChain and LangGraph, using prompt engineering and retrieval-augmented generation over client data.
- Scoped and negotiated requirements directly with non-technical clients, managing expectations, timelines and trade-offs without an intermediary project manager.
- Used agentic AI coding tools (Claude Code, Codex) across the development lifecycle to accelerate delivery without compromising code quality.

Software Engineer — ServiceRocket, Kuala Lumpur June 2021 – October 2023

Associate Software Engineer, June 2021 – June 2022; promoted to Software Engineer, June 2022 – October 2023.

- Owned the full development lifecycle of enterprise Confluence DC/Server applications for global customers, and prototyped products across Confluence Cloud, Workplace by Meta, Miro, Jira and Discord.
- Built integrations with Google, Salesforce and Jira, and automated cross-platform workflows using agentic, multi-step orchestration patterns to reduce manual intervention.
- Implemented CI/CD pipelines with GitHub Actions and Jenkins, improving deployment speed and stability across the team.
- Resolved complex production issues directly with support teams and clients, communicating root cause clearly to technical and non-technical audiences under time pressure.
- Mentored junior engineers through onboarding, code review and architectural guidance, raising code quality and shortening ramp-up time for new joiners.

Fullstack Developer — dJava Factory, Kuala Lumpur November 2019 – March 2020

- Full-stack web and mobile development on an internal product team, working to shared standards and peer review.

SELECTED PROJECTS

- **Frontier** — Funded by Bluedot Impact's Rapid Grant; research harness testing whether model compression (quantisation, distillation) degrades calibration faster than accuracy; ECE and Brier implemented from scratch with bootstrapped confidence intervals. (PyTorch, Hugging Face, vLLM)
- **Glosward** — grounded, cited RAG assistant over Hugging Face documentation, with a QLoRA-tuned adapter that reshapes tone without touching retrieved facts; hit@5 0.923, MRR@5 0.740 on a 26-question evaluation set. (PyTorch, PEFT, bitsandbytes, Chroma, Gradio)
- **Agentic-DS** — autonomous, offline data-science agent that profiles an unseen CSV, plans preprocessing and models from the data's own properties, then reflects and replans until convergence; changes kept only if significance-tested. (Python, scikit-learn, pandas, imbalanced-learn, pytest)
- **AuthLedger** — TypeScript identity and payments backend: hand-written session auth, TOTP MFA and OAuth gating a Stripe-integrated, database-enforced double-entry ledger; 25 test files run against a real Postgres instance. (Fastify, Kysely, Postgres, Terraform)

- **Tombstone** — from-scratch CRDT (Replicated Growable Array) for real-time collaborative text editing, with convergence proven by property-based tests against an independent reference oracle. (TypeScript, fast-check, WebSocket)

KEY SKILLS

Programming: Python, TypeScript, Java, Spring, JavaScript/Node.js

Generative AI / LLMs: Prompt engineering, retrieval-augmented generation (RAG), embeddings and vector search, function/tool calling, structured outputs, LLM APIs, PyTorch, Hugging Face Transformers, vLLM

Fine-Tuning / PEFT: LoRA, QLoRA, Hugging Face PEFT, adapter configuration, 4-bit NF4 quantisation with bitsandbytes, instruction/SFT dataset construction, adapter merging and hot-swapping, adapter-vs-base evaluation

AI / ML: Machine learning, deep learning, model evaluation and calibration (ECE, Brier score), explainability, neural networks, natural language processing, data science

Agentic Systems: LangChain, LangGraph, agent frameworks, multi-step tool use, workflow orchestration, Model Context Protocol (MCP), AI-assisted coding (Claude Code, Codex)

Backend / Data: PostgreSQL, SQL, Fastify, REST APIs, database design, data modelling

Cloud & DevOps: AWS, Azure, Docker, Terraform, GitHub Actions, Jenkins, CI/CD, Git version control

Practices: Agile, Scrum, software development lifecycle (SDLC), system design, API integration, responsible AI, automated testing, code review

EDUCATION

MSc Artificial Intelligence — University of Essex, Colchester — October 2025 – October 2026

- ***Thesis***: a calibrated, explainable retrieval-augmented assistant for self-management of mental health in healthcare workers. Researching the signals that portray calibration, explainability, and safety, and how they correlate to the usefulness and perceived reliability and safety of the system.
- **Neural Networks and Deep Learning**: studied convolutional and recurrent neural networks and transformer architectures, covering backpropagation, gradient descent optimisation, regularization, and transfer learning.
- **Machine Learning**: covered supervised and unsupervised learning, feature engineering, model selection and cross-validation, ensemble methods, and evaluation metrics.
- **Data Science and Decision Making**: applied statistical inference, exploratory data analysis, data wrangling and visualisation, and predictive modelling to support evidence-based decision-making under uncertainty.
- **Intelligent Systems and Robotics**: explored search and planning algorithms, reinforcement learning, computer vision and sensor fusion, and multi-agent systems for autonomous decision-making.

BSc Computer Science (First Class Hons) — Asia Pacific University, Kuala Lumpur — February 2018 – May 2021

- **Networking and Security**: studied TCP/IP networking protocols and network architecture alongside cryptography, authentication, threat modelling and secure system design.
- **Concurrent Programming and Embedded Systems**: implemented multithreaded and parallel algorithms in Java and C, covering synchronisation, race conditions and deadlock avoidance.
- **Computer Vision and Object-Oriented Programming**: applied image processing, feature detection and object recognition, and built maintainable systems using OOP principles, SOLID and design patterns in Java.
- **Cloud Application Development and Blockchain**: designed and deployed distributed cloud applications covering virtualisation, containerisation, scalability and cloud service models, and built distributed ledger applications using smart contracts and consensus mechanisms.

CERTIFICATIONS

- **Professional Scrum Developer (PSD)** — September 2022.
- **Technical AI Safety, BlueDot Impact** — in progress, certificate expected September 2026.

PROFESSIONAL STRENGTHS

- **Communication & stakeholder management** — translates technical trade-offs for non-specialist and client audiences.
- **Mentoring & knowledge sharing** — onboards junior engineers and gives constructive, specific code review.
- **Collaboration & teamwork** — cross-functional, distributed delivery in Agile/Scrum environments.
- **Ownership & accountability** — takes work from ambiguous requirements through to production release.
- **Analytical problem-solving & attention to detail** — structured debugging, automated testing, evidence over assumption.
- **Ethical judgement & responsible AI** — prioritises calibration, explainability and safety in high-risk contexts.